

ERNAGE, Belgium

WMO: 064590

Lat: 50.5819N

Lon: 4.6892E

Elev: 515

StdP: 14.42

Time Zone: 1.00 (EUC)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 18.6	(c) 23.0	(d) 14.1	(e) 11.5	(f) 19.4	(g) 20.7	(h) 15.8	(i) 24.7	(j) 29.4	(k) 47.5	(l) 27.1	(m) 45.2	(n) 7.0	(o) 60	(p) 0.576

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 18.0	(c) 84.6	(d) 69.3	(e) 80.3	(f) 67.4	(g) 76.8	(h) 65.7	(i) 71.5	(j) 80.7	(k) 69.2	(l) 77.0	(m) 67.2	(n) 74.3	(o) 7.9	(p) 190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.5	106.6	75.3	66.6	99.8	73.1	64.6	93.1	70.4	35.6	81.2	33.6	76.8	32.0	74.1	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 23.4	(b) 20.7	(c) 18.3	(e) 13.0	(f) 92.2	(g) 9.6	(h) 4.5	(i) 6.1	(j) 95.5	(k) 0.5	(l) 98.1	(m) -4.9	(n) 100.7	(o) -11.9	(p) 104.0		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			12.8	75.0	9.7	2.8	5.8	77.0	0.1	78.7	-5.3	80.3	-12.3	82.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	50.5	36.9	37.7	43.6	49.0	55.3	60.7	64.5	63.8	58.2	51.7	44.2	39.3	(6)
(7)		DBStd	11.58	7.77	7.51	6.53	6.43	6.56	5.63	5.81	4.96	5.02	6.33	6.33	7.18	(7)
(8)		HDD50	1666	406	345	215	94	21	0	0	0	1	56	193	335	(8)
(9)		HDD65	5452	870	765	665	479	308	156	81	81	214	414	623	796	(9)
(10)		CDD50	1837	1	0	15	65	185	320	449	427	245	107	20	3	(10)
(11)		CDD65	150	0	0	0	0	7	25	65	43	9	1	0	0	(11)
(12)		CDH74	1469	0	0	0	9	69	251	620	436	81	3	0	0	(12)
(13)		CDH80	404	0	0	0	0	9	56	186	137	16	0	0	0	(13)
(14)	Wind	WSAvg	9.0	10.6	10.3	10.2	8.6	8.2	8.0	7.8	7.7	7.9	8.4	9.6	10.7	(14)
(15)	Precipitation	PrecAvg	31.8	2.5	2.1	2.7	2.2	2.8	3.1	2.7	2.7	2.4	2.7	3.0	2.9	(15)
(16)		PrecMax	43.2	5.8	4.3	5.5	5.2	5.5	6.3	6.7	5.7	6.5	6.3	5.0	7.3	(16)
(17)		PrecMin	17.9	0.9	0.3	0.7	0.5	0.7	0.4	0.9	0.7	0.6	0.2	0.9	0.5	(17)
(18)		PrecStd	5.8	1.2	1.1	1.3	1.2	1.3	1.5	1.5	1.3	1.4	1.7	1.0	1.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.8	58.2	66.0	75.0	80.0	86.7	91.4	90.6	82.2	73.2	62.4	55.7	(19)
(20)			MCWB	52.4	47.1	54.9	58.6	66.3	70.3	70.0	69.8	68.6	61.9	59.3	53.4	(20)
(21)		2%	DB	51.8	51.7	61.6	69.7	75.3	80.4	84.7	83.7	75.5	67.6	58.9	53.4	(21)
(22)			MCWB	50.6	48.5	52.8	57.5	62.7	68.0	70.1	68.8	64.6	62.1	56.6	51.2	(22)
(23)		5%	DB	49.1	49.6	57.1	64.6	71.5	76.2	80.3	78.7	71.1	63.6	56.3	51.3	(23)
(24)			MCWB	47.9	47.5	51.4	55.7	61.7	65.9	67.2	66.9	63.1	59.4	54.4	49.9	(24)
(25)		10%	DB	46.9	47.6	53.2	60.3	67.6	72.4	76.7	74.3	67.2	60.8	53.7	49.0	(25)
(26)			MCWB	45.8	45.9	48.8	53.3	60.2	64.2	66.1	65.0	60.6	58.1	52.1	47.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.2	50.9	56.3	61.6	69.1	73.0	74.5	74.4	69.5	65.1	60.3	54.4	(27)
(28)			MCD B	53.5	53.4	62.6	71.1	76.3	82.7	84.3	85.3	80.0	70.3	61.1	55.2	(28)
(29)		2%	WB	50.7	49.1	54.0	58.5	65.3	69.6	71.9	70.5	66.2	62.4	56.8	51.9	(29)
(30)			MCD B	51.6	50.7	60.4	66.3	71.3	76.6	81.6	78.5	73.2	65.9	58.2	52.9	(30)
(31)		5%	WB	48.0	47.6	51.7	56.2	62.8	67.2	69.6	67.9	64.0	60.2	54.7	50.0	(31)
(32)			MCD B	48.9	49.2	56.2	62.8	68.6	74.3	77.9	74.8	68.7	63.0	56.3	51.3	(32)
(33)		10%	WB	45.8	45.8	49.6	54.0	60.7	64.9	67.3	66.0	62.0	58.3	52.4	47.9	(33)
(34)			MCD B	46.7	47.7	52.7	59.9	66.7	71.4	74.6	73.0	66.0	60.7	53.6	48.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.8	9.1	13.1	16.5	16.9	17.6	18.0	17.3	15.6	12.6	9.1	7.6	(35)
(36)			MCDBR	8.6	12.7	20.6	23.1	22.6	24.0	25.0	25.6	21.4	16.5	11.9	9.0	(36)
(37)			MCWBR	7.9	9.7	13.3	13.2	11.8	13.2	11.9	12.4	11.7	11.0	9.8	7.7	(37)
(38)		5% WB	MCD BR	8.6	11.3	18.1	20.3	19.0	21.2	22.2	21.0	18.2	14.9	11.0	8.8	(38)
(39)			MCWBR	8.0	9.1	12.4	12.2	11.5	12.7	12.0	12.0	11.4	10.5	9.4	8.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.319	0.341	0.374	0.409	0.419	0.416	0.419	0.411	0.382	0.357	0.332	0.316	(40)	
(41)		taud	2.412	2.375	2.286	2.205	2.218	2.245	2.239	2.285	2.353	2.419	2.414	2.381	(41)	
(42)		Ebn at Noon	226	250	260	263	266	267	264	260	255	240	217	206	(42)	
(43)		Edh at Noon	20	26	34	41	43	42	42	38	32	25	20	18	(43)	
(44)	All-Sky Solar Radiation	RadAvg	247	460	826	1310	1556	1695	1611	1372	1037	613	292	195	(44)	
(45)		RadStd	36	91	117	152	156	154	193	146	109	51	35	30	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (77 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page